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A Multiple Imputation Workflow for Handling Missing Covariate Data in Pharmacometrics Modeling

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ABSTRACT

Covariate missingness is a prevalent issue in pharmacometrics modeling. Incorrect handling of missing covariates can lead to biased parameter estimates, adversely affecting clinical practice and drug development dosing decisions. Single imputation is usually favored by pharmacometricians for its simplicity, but it ignores the uncertainty about imputed values, potentially leading to biased estimates and standard errors. Multiple imputation, in contrast, generates multiple plausible values from a predictive distribution, addressing this uncertainty and thus is a preferable approach over single imputation to handle covariate missingness. Yet, its application in pharmacometrics remains limited due to perceived complexity. To address this, we developed a multiple imputation workflow specifically tailored for pharmacometricians, encouraging wider adoption of this more reliable method in pharmacometrics modeling. We compared single imputation and multiple imputation in estimating covariate effects using a publicly available dataset on warfarin pharmacokinetics in healthy volunteers. A one-compartment population pharmacokinetic model with baseline body weight as the only covariate was used to describe the warfarin pharmacokinetics. We simulated five scenarios in which 6.25%, 12.5%, 25%, 50%, and 75% of the subjects had their body weight missing under a missing at random mechanism conditioned on age and sex. We confirm that multiple imputation better reflects uncertainty estimates than single imputation, regardless of the degree of missingness. This confirms multiple imputation as a superior alternative to single imputation for handling missing covariate data in pharmacometrics.

1 | Introduction

Missingness of covariate data is a prevalent challenge in medical research, including pharmacometrics [1, 2]. Inappropriate handling of covariate missingness is detrimental as it can adversely affect drug development and clinical practice decisions [3, 4]. At worst, it can have negative consequences for patients, including systematic over- or underdosing, thus highlighting the critical importance of handling missing covariate information accurately in pharmacometrics [3, 5].

Covariate missingness can be classified into three missing data mechanisms: missing completely at random (MCAR), missing

at random (MAR), and missing not at random (MNAR) [6]. The mechanism of covariate missingness can be classified based on whether the missing data depends on observed and/or unobserved data, including the outcome or other covariates [7, 8]. In the case of MCAR, missingness is assumed to be entirely random, independent of both observed and unobserved data, as in cases where a serum sample accidentally gets lost [6]. With MAR, covariate missingness depends on observed data but not on unobserved data [6]. For instance, in a population pharmacokinetic (popPK) model, serum creatinine (SCr) may be missing more often in people with a higher body mass index (BMI). In MNAR, missingness depends on unobserved data, illustrated by higher missingness of BMI in people with obesity [9].

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Commonly suggested methods for handling missing covariate data include single imputation (SI) and multiple imputation (MI) [2]. In SI, missing values are imputed once, resulting in a seemingly true, complete dataset, but substantially disregarding the uncertainty about the imputed values [1, 2]. On the other hand, MI involves inference based on multiple analyses from datasets in which missing data are replaced with more than one value, explicitly acknowledging the uncertainty about the values that were not observed. MI was shown to be superior to SI, especially under MAR [8].

Despite being a preferred method for handling missing data, the application of MI in pharmacometrics remains limited [2]. Pharmacometricians still favor SI over MI due to its simplicity and ease of communication with non-modelers [6]. Another reason for this preference may be the lack of tailored MI guidance for pharmacometricians, hindering the implementation of MI in pharmacometrics.

To address this gap, we present a comprehensive workflow for handling missing covariate data using MI, specifically focusing on pharmacometrics in this tutorial. Our goal is to encourage the wider adoption of MI in the field. We demonstrate the superiority of MI over SI through a case study involving the popPK of warfarin in healthy volunteers [10].

2 | Simple Methods for Handling Missing Data

2.1 | Complete Case Analysis

Complete case analysis (CCA) was the most common method to handle missing covariate data due to its convenience [11]. Also known as listwise deletion, this method involves removing subjects with incomplete covariate data and modeling only those with complete covariate data [6]. It is valid and thus produces valid inferences when the covariate missing mechanism is MCAR [11]. It is also the default way of missing data handling in many statistical packages, including Phoenix WinNonlin [6, 12].

However, there are several issues associated with CCA. First, since MCAR rarely happens in practice, the parameter estimates will likely be biased [13]. Second, the reduced sample size decreases the statistical power, rendering the parameter estimates less precise [11]. Finally, CCA can lead to inconsistent reporting as different analyses often use different subsamples from the same overall sample [11]. This makes it challenging to compare between analyses and to generalize to the study population [11]. The Committee for Medicinal Products for Human Use (CHMP) typically does not endorse CCA because it violates the intent-to-treat principle [14].

2.2 | Single Imputation

In SI, a single value is used to replace each missing data point. This method is also referred to as “naive substitution” in which either the median or mean value is often used to substitute the missing values [6]. For instance, missing SCr values can be replaced with the sample mean or median of individuals

with observed SCr. Despite being a simple and fast method for missing covariate data handling, mean or median imputation reduces variability estimates because it treats imputed values as if they were truly observed, ignoring the uncertainty associated with the missing observations. This results in falsely overestimating the precision of the results [8] and in an increased risk of committing the Type 1 error of identifying a significant covariate effect when it, in fact, does not exist [13]. Furthermore, mean or median imputation disturbs the relations between variables in the sample [11]. For example, older subjects may have higher SCr than younger ones on average. Such a correlation between age and SCr is considered by neither imputation. Only the mean or median of the marginal distribution of SCr is preserved in this case, which is unbiased only in the case of MCAR [15]. This approach is generally recognized as doing more harm than good. At best, it contributes no additional information when the missing data is minimal (less than a few percent) [6].

Other SI methods very commonly used in pharmacometrics are the last observation carried forward (LOCF) and the next observation carried backward (NOCB), where the closest available observed covariate value is taken forward or backward to fill the holes in the dataset for that subject [8]. LOCF and NOCB are commonly applied for imputing time-varying covariates for subjects with at least one covariate observation available [6]. For example, suppose that SCr was measured only on days 1 and 7, while PK samples were collected daily. Hence, there is no SCr concentration available on days 3–6. LOCF and NOCB mean that the SCr concentration on days 3–6 is set to the concentration on day 1 and 7, respectively. Consequently, it is assumed that the covariate remains unchanged over time, risking an artificial reduction of the variability [6]. NOCB is the default in NONMEM used to interpolate covariate values within the interval between observations [16]. Similar to mean or median imputation, LOCF and NOCB yield implausible imputations as both methods ignore the time-varying nature of the covariates, which is problematic unless there is evidence that they are constant within a certain time interval [8]. Consequently, LOCF yields bias in both directions for all parameter estimates even under MCAR [13].

The SI techniques discussed are still widely used in pharmacometrics due to their ease of implementation and convenience in communication with non-modelers [6]. However, as explained before, they share common flaws of not considering the uncertainty in the estimate of the imputed value and being likely to result in bias [6, 8]. In fact, SI is shown to result in seriously biased and imprecise parameter estimates when utilized to handle missing covariates in popPK modeling [2]. It is thus crucial for pharmacometricians to challenge the status quo and adopt methods that address these limitations of SI. One such method is MI, which will be explored in detail below.

3 | Multiple Imputation of Missing Covariate Data

MI was first presented by Rubin in the late 1970s as a tool to overcome the limitations of the conventional missing data handling methods [17]. Specifically, it solves the problem of shrunk standard errors due to overly confident SI [6, 8]. The

concept of MI is based on replacing each missing value with multiple plausible values—instead of a single value—thereby reflecting the uncertainty about the missing data [6, 8]. Moreover, in addition to expectation–maximization and direct likelihood methods, MI is valid under MAR, which is a more realistic assumption as compared to MCAR [8, 18]. In MAR, the distributions of the observed and missing data of a certain covariate are the same when conditioned on other observed variables. Therefore, the idea of MI is to build a model using observed data to impute missing data [19]. As in the SCr example above, we can impute missing SCr values using an imputation model with age as a predictor. Another advantage of MI is that it helps preserve the relationships, including the correlations between variables in the sample [15]. It is now considered the best method for dealing with incomplete data in many fields. In pharmacometrics, it has been shown to yield less biased and more precise population parameter estimates compared to CCA and SI methods. Nevertheless, its adoption in the field is uncommon [1, 2].

An MI workflow generally consists of five main steps (Figure 1). The first two steps, “specification of the imputation method” and “selection of variables to be included in the imputation model,” involve building a model for each incomplete covariate to perform MI [8]. In the third step, the “creation of M multiply imputed datasets,” several M complete datasets are created by replacing missing covariate values with plausible values [11]. After that, an independent analysis is conducted for each imputed dataset to estimate model parameters [11]. This is a relatively straightforward step in which the standard modeling workflow is applied to each dataset as if it were complete. Lastly, the M parameter estimates are pooled into one point estimate along with its variance. These results allow making valid inferences about the data while accounting for the uncertainty about the missing data and having the correct type 1 error rate [11].

Below we discuss the considerations needed for performing MI in the order of the general workflow. In this discussion, we assume that all variables are fully observed except for covariates and that covariate data missingness is MAR, which is overall a reasonable assumption in PK and pharmacodynamic modeling.

3.1 | Preliminary Step: Exploration of Missingness Mechanisms and Patterns

Generally, there is no formal test to check whether data missingness is MAR, while there are measures to test the MCAR assumption. Specifically, we can do so by separating data into two groups, one with missing and the other with observed covariate information, and then comparing characteristics between these two groups [8, 20]. This can be done via statistical tests such as t -tests, Chi-square tests, or Little’s MCAR tests [20, 21], which are available in the *finalfit* and *nanian* packages in R [22, 23]. If the missing data mechanism is likely MCAR, these characteristics should not differ significantly [15].

It is also useful before imputing to gain insights into missing data patterns of the pharmacometrics dataset, which tells us

where the missing values are taking place. This is because the missing data pattern influences the amount of information that can be transferred between variables and the precision of imputation. Overall, the pattern of data missingness can either be classified as univariate versus multivariate or monotone versus non-monotone. The univariate pattern happens when there is only one variable with missing data [24]. In practice, the multivariate pattern is more commonplace where more than one covariate has missing data. On the other hand, a monotone missingness pattern arises when the missingness of covariates can be ordered in a way that if one covariate is missing then all subsequent covariates are also missing. This can occur when a patient drops out of the study, thus all the subsequent measurements of their covariates are not available. The univariate pattern represents a special case of the monotone pattern, and significant computational efficiencies can be achieved when dealing with monotone data [20]. Similar to the univariate pattern, the monotone pattern is not commonplace in practice, and what pharmacometricians usually encounter is a non-monotone missing pattern [20].

3.2 | Step 1: Specify the Imputation Method for Each Incomplete Covariate

It is necessary to differentiate between the imputation model and the analysis model. The imputation model only serves the purpose of imputing missing data to provide reasonable imputations and is thus not of direct interest. On the other hand, the analysis model is the one that is of ultimate scientific interest and is the focus of the research question, in the case of pharmacometrics, often a popPK model.

3.2.1 | Method 1: Parametric Imputation Methods

The most intuitive way to generate imputations for a certain covariate is to use an imputation model built on observed covariate and outcome values. The type of model depends on the missing covariate data type (Table S1). For instance, a linear regression model is appropriate when the missing covariate is continuous by nature, such as SCr and body weight; a logistic regression model can be an option for binary missing covariates like biological sex, while a proportional odds model is suitable when the data is ordinal, as in the case of the sequential organ failure assessment (SOFA) score. After fitting an imputation model, there are several options on how missing values can be imputed. One of them is to directly impute the deterministic predictions from the imputation model. This approach is not suitable for MI, since it does not take into account the uncertainty about the missing covariate, thus generating the same values with each imputation. Such an approach can be preferably improved by including parameter uncertainty using either Bayesian or bootstrap methods [8]. These two methods are available in many R functions (Table S1).

3.2.2 | Method 2: Predictive Mean Matching

One of the issues of the fully parametric methods is that they can generate implausible values. For example, covariates that can

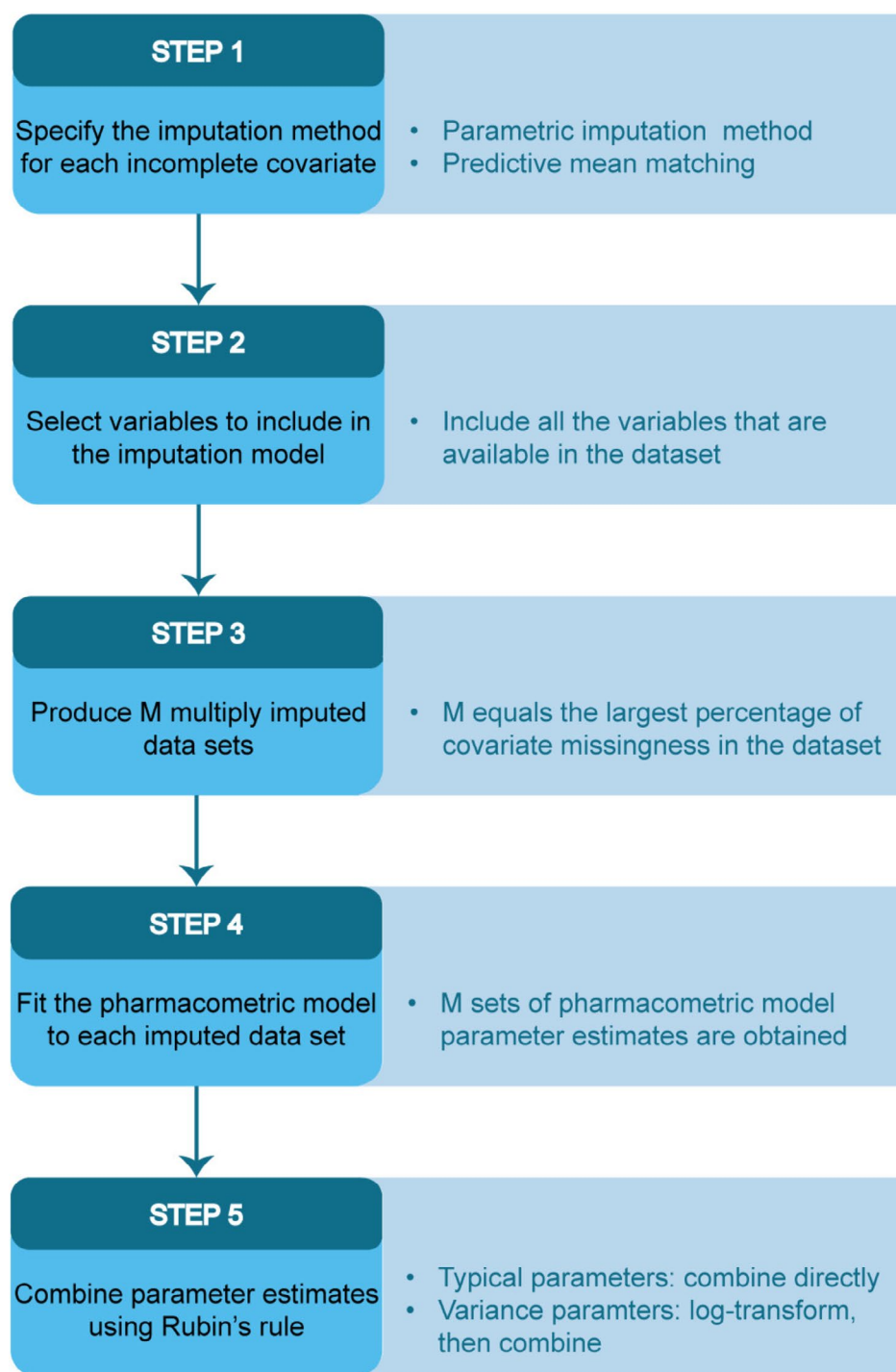


FIGURE 1 | A general workflow of multiple imputation.

only be positive (e.g., SCr, body weight, ...) may have imputed values that are negative. This typically occurs when model assumptions, such as normally distributed residuals in a linear regression model, are violated. One option is to transform the variable before imputation to meet the normality assumption. For instance, log-transforming SCr before imputation and then back-transforming it to the original scale can help avoid negative values [11]. A second option is to draw imputations from the observed values by using predictive mean matching (PMM), which can also be used to impute both numerical and categorical data [19].

Generally, PMM predicts the value of target variable Y using the specified imputation model and indirectly uses these predictions for imputing the missing values. Essentially, this technique can be considered a “smart way” to sample missing values from the observed ones with replacement. Specifically, if Y is continuous, PMM first calculates the predicted value of the covariate of interest Y by fitting a linear regression model between Y and the predictors [25]. For each subject with missing data on Y , PMM identifies a set of candidate donors from subjects with no missing data on Y (typically five members) [26]. These potential donors have linear predictors that are closest to the predicted value for

TABLE 1 | Pros and Cons of parametric and predictive mean matching multiple imputation methods.

Methods	Parametric imputation	Predictive mean matching
Pros	Can be used even when the sample size is small (≤ 5).	Always generate plausible values Preserves the location of quantiles More reliable to imputation model misspecification
Cons	May generate implausible values Susceptible to imputation model misspecification Do not preserve the location of quantiles	May impute one value many times if the sample size is small (≤ 5). Cannot be used to extrapolate beyond the range of the data, or to interpolate within the range of the data if the data at the interior are sparse

the missing entry [26]. One donor is randomly selected from all the candidates, and the observed value of the donor is then used to replace the missing value [19]. This results in the same data range between imputed and observed values. Moreover, as it does not fully rely on the imputation model to generate imputations, PMM is a more robust method against model misspecification [27]. Because of these advantages, PMM is the default method for MI of continuous variables in the *mice* package in R [15].

However, this method has several associated pitfalls, such as the duplication of the same donor value many times, which can artificially and incorrectly reduce the variability in the data and might introduce bias [8]. This issue occurs more likely when the size of the donor pool is small or if there are many more missing data than observed data. Furthermore, the commonly employed linear regression imputation model in PMM might not be appropriate when there is a presence of strong nonlinear relations of the missing covariate with other variables. Therefore, although having some great properties as described, PMM is not a substitution for careless modeling. Efforts should be spent on correctly specifying the imputation model rather than expecting PMM to do the work since model misspecification can lead to poor imputation performance [8, 28]. We presented the pros and cons of parametric and predictive mean matching multiple imputation methods in Table 1.

3.2.3 | Method 3: Multiple Imputation by Fully Conditional Specification

All the above techniques apply to the context of univariate missing covariates. In the case of the more frequently encountered multivariate missing covariates, fully conditional specification (FCS), also known as chained equations, is usually employed to address the issue [19]. FCS is a strategy involving specifying an imputation model for each incomplete variable based on all other variables and creating imputations per incomplete variable iteratively [29, 30].

One way of implementing FCS in performing MI is called the multivariate imputation by chained equations (MICE) algorithm [8]. The algorithm is described below.

1. Specify an imputation model for each of the k covariates with missing values.

2. For each of the k missing covariates, fill in the missing values with random draws from subjects with observed values for the respective covariate. These initial imputed values only serve as placeholders and will be replaced by better imputed values in the following steps of the algorithm.
3. For the first covariate with missing values:
 - a. Regress this first covariate on all the other variables using those subjects with complete data on the first covariate and observed or currently imputed values of the other variables.
 - b. The estimated regression coefficients, their variance-covariance matrix (and the estimated variance of the residual distribution in case a linear regression model was fit for a continuous covariate) are extracted from the regression model estimated in (a).
 - c. Randomly draw a random set of parameters in (b) from their posterior distribution.
 - d. Use the set of randomly selected parameters in (c) to impute values for subjects with missing data on the first covariate using the fitted model.
4. Repeat step 3 for the remaining $k-1$ missing covariates. Steps 3 and 4 form 1 iteration of the imputation process resulting in 1 imputed dataset.
5. Repeat steps 3 and 4 the desired number of times (suggested 5–20 iterations). The set of imputed values at the last iteration is used as the imputed values in the first imputed data set.
6. Repeat steps 2–5 $M-1$ times to produce M imputed data sets (the choice of M , the number of imputed data sets, is discussed in the section “Produce M multiply imputed data sets”).

MICE is a very flexible algorithm as it allows users to specify different models for different missing covariates in step 1. Every method that was described above in the case of univariate missing covariate can be used in this step. After having filled in missing values with placeholders drawing from observed subjects in step 2, the algorithm repeats steps 3 and 4 for several iterations to create one imputed dataset in step 5. The required number of iterations is generally low, in which 5–20 iterations are usually enough for the algorithm to converge. A higher number is needed in cases

with high correlations between covariates or when the rates of missing covariates are high [8, 15]. The imputed values acquired during the final iteration serve as the imputed values for the first imputed dataset. The whole process is then repeated for another $M - 1$ times to generate M multiply imputed datasets (step 6).

3.3 | Step 2: Select Variables to Include in the Imputation Model

The proper imputation model should not contradict the pharmacometrics analysis model [31]. This means the imputation model should include all the variables present in the analysis dataset; otherwise, the pharmacometrics analysis risks biased covariate estimates [11]. Additionally, including all the variables helps reflect the correlation between variables in the imputation model. Moreover, these variables need to be incorporated correctly. Theoretically, this means that if the pharmacometrics model includes an interaction term, this term must also be in the imputation model to avoid having a biased interaction estimate [32]. However, such an interaction term is commonly only introduced during model development, undermining practical implementation.

Importantly, the outcome variable must be included in the imputation model [19]. For instance, in handling missing covariates in a popPK model, drug concentrations at all time points should be included in the imputation model, as they carry information about the missing covariates [33]. Omitting this could artificially weaken the association between covariates and drug concentration [34]. Additionally, it is beneficial to include variables predictive of the incomplete covariates in the imputation model [19]. This approach has two main benefits: first, it supports the MAR assumption, reducing bias [35], and second, it improves the quality of imputations, thus enhancing the precision of the analysis model parameter estimates [19].

3.3.1 | Consideration: How to Handle Derived Covariates?

Performing MI can pose challenges when dealing with missing derived covariates, which are computed from other covariates. Examples commonly encountered by pharmacometricians include BMI (derived from height and weight) and estimated glomerular filtration rate (eGFR, derived from age, sex, and serum creatinine) [36, 37]. These covariates become absent if at least one of their components is unavailable for computation. Consequently, three primary options exist for imputing these variables.

The first option “impute, then transform”, involves imputing the missing component covariates first and then calculating the derived covariates afterward [31]. For instance, if height or weight is missing, these are imputed first, and BMI is then computed. This approach preserves the relationship between the derived covariate and its components but may introduce bias if the derived covariate itself is not included in the imputation model [31].

The second option, “transform, then impute”, or “just another variable”, treats the derived covariate as an independent variable in the imputation model [15]. Here, the derived and component covariates (e.g., BMI, height, and weight) are imputed

together. This approach aligns the imputation model with the analysis model and has been shown to perform the best in estimating interaction and quadratic effects in regression models [31, 38]. A drawback is that it can disrupt the relationship between derived covariates and their components, leading to implausible values [8].

The third technique is called “passive imputation,” or “transform, impute, then transform again.” It calculates the transformation on the fly within the imputation algorithm. This incorporates the derived covariate into the imputation model, preserving the relationship between the derived and component covariates [31]. Although it intuitively makes sense, von Hippel showed that “passive imputation” is merely a more sophisticated version of the “impute, then transform” technique, and both methods share the same biases when covariates are MAR [31]. Bartlett et al. also demonstrated severely biased covariate effect estimates obtained from “passive imputation” when covariates are MAR [39].

All three methods discussed above have limitations, either producing implausible imputed values or introducing high biases, highlighting the need for improved techniques. Substantive model-compatible FCS has emerged as a promising alternative, showing substantial improvements over the other methods when imputing squared terms and interactions [39]. However, as it remains relatively new and underexplored, this approach represents a promising area for further research [8].

Example R codes showcasing these three methods are available in the Data S1.

3.4 | Step 3: Produce M Multiply Imputed Data Sets

One of the most important considerations in MI is the number of imputations M [11]. It is conventionally recommended that three to five imputed data sets are adequate for the amount of missing covariates up to 50% [40]. This early recommendation was based on the efficiency argument, which compares the variance of parameter estimates when imputing M times versus the infinite number of times. However, this approach does not ensure the reproducibility of results across repeated MI analyses [19, 41]. This means that two pharmacometricians can report substantially different estimates despite having the same M and using the same dataset [41]. Therefore, it is essential to make sure that the standard error estimate is replicable, which requires a larger M .

A higher M is always better, but it is important to take the computational burden of analyzing all these M imputed datasets into account [8]. A fairly practical rule of thumb for M is that it should be at least equal to the proportion of individuals with missing covariate data [19]. This has been shown to result in minimally variable parameter estimates and standard errors across repeated MI analyses [11]. Recently, von Hippel proposed a two-stage method for determining M , based on the quadratic relationship between M and the degree of missingness [41]. Additionally, Nassiri et al. introduced a framework in which is determined by comparing the estimates under M and $M + 1$ imputations, ensuring that the difference between these estimates falls below a pre-defined threshold [42]. Although these may be statistically sound

methods, M obtained from them can be highly variable [41, 42]. With the computational power available nowadays, it is common to have between 20 and 100 imputed data sets [11].

3.5 | Step 4: Fit the Pharmacometric Model to Each Imputed Data Set

After having generated M complete datasets, the substantive pharmacometrics analysis of interest is carried out for each complete data set individually. Such an analysis is the same as what would be done without any missing values. Thus, if the analysis model is a popPK model, there will be M popPK models fitted simultaneously. The population parameters of interest (e.g., clearance, volume of distribution) and their standard errors are then extracted from the results of each popPK analysis.

3.6 | Step 5: Combine Parameter Estimates Using Rubin's Rule

Once the population parameters of interest (including both fixed-effect and random-effect parameters) as well as their standard errors have been estimated in the M imputed data sets, they will be combined using Rubin's rules [43]. Let $\theta^{(i)}$ denote the estimated parameter of interest (e.g., clearance) obtained from the analysis in the i th imputed data set ($i = 1, \dots, M$). The pooled clearance estimate of the M imputed datasets is $\theta = \frac{1}{M} \sum_{i=1}^M \theta^{(i)}$. The MI estimate of the population parameter is simply the average value of the estimated population parameter across the M imputed data sets [19].

Calculating the variance of the estimated population parameter is more complicated. This involves computing two components, which account for the uncertainty of the estimated population parameter per imputation and the variation of such parameter between imputations [19]. Let $W^{(i)}$ denote the estimated variance of $\theta^{(i)}$; then, the within-imputation variance of θ is defined as $W = \frac{1}{M} \sum_{i=1}^M W^{(i)}$. This is the average of the estimated variance of the population parameter estimate throughout the M imputed data sets. On the other hand, the between-imputation variance of θ , denoted by B , is calculated as $B = \frac{1}{M-1} \sum_{i=1}^M (\theta^{(i)} - \theta)^2$ [19]. B represents the degree to which the population parameter estimate varies across the M imputed data sets [19]. W and B can be combined using Rubin's rule to calculate the estimated variance of θ as $\text{var}(\theta) = W + [1 + M^{-1}]B$ [8]. This quantity reflects both the average variability within and between imputations in θ .

After obtaining the estimated θ and $\text{var}(\theta)$, one can make inferences about the population parameter by calculating its 95% confidence interval (CI). This 95% CI is computed based on the Student's t -distribution of θ with ϑ degrees of freedom, which can be derived based on the formula below [44].

$$\vartheta = \frac{1}{(M-1)} \left[\frac{(1+M^{-1})B}{\text{var}(\theta)} \right]^2$$

It is important to note that Rubin's rule is only valid for parameters that are approximately normally distributed [19]. Examples

of such parameters in pharmacometrics are typical PK parameters such as clearance or volume of distribution. For non-normally distributed random-effect parameters, such as inter-individual variability (ω^2), or residual variability (σ^2), a transformation to a normal distribution, for instance using log transformation, is required before Rubin's rule can be applied [19].

3.7 | Multilevel Multiple Imputation

In pharmacometrics, missing covariates are often of a longitudinal and hierarchical (multilevel) nature. Specifically, these covariates are clustered into groups due to repeated measurements within each group. For instance, SCr was measured repeatedly over time within the same patients. These data are clustered (or correlated) because assessments of SCr within the same patient are more similar to each other than assessments of SCr between different patients [7, 20]. In this context, the appropriate conditional imputation model for SCr when employing the FCS strategy is a mixed-effects model, with the cluster variable being the patient identifier. In addition, predictors that vary within a cluster are called the level 1 predictors, while predictors that are constant within the cluster are called the level 2 predictors [8]. In the SCr example, a level 1 predictor can be albumin concentration, which varies within patients, and body weight can be a level 2 predictor.

Performing MI in this context is more challenging than in single-level contexts described before for two main reasons. First, not only is the variable of interest (SCr) but also variables of different levels can be missing, which has different implications for the imputation process [8]. Second, missing covariates within each cluster can either be partly (i.e., sporadically) or completely (i.e., systematically) missing, which also affects the choice of the imputation model [45]. Additionally, the complexity of a multilevel imputation task is elevated by other factors such as whether or not error variance differs across clusters (heteroscedasticity vs. homoscedasticity) [45]. There is currently no one method that can address all the issues of such a task, and MI employing mixed-effects models is still under development [8].

Nevertheless, it has been shown that ignoring the clustering and imputing the data by a single-level method can be hazardous, as it can bias the model's parameter estimates [46]. Thus, it is recommended to avoid single-level imputation except for when there are a few cases with missing data (e.g., < 5%) and the intra-class correlation is low (e.g., < 0.1) [46]. The methods available in R mostly employ two-level imputation [8]. R functions for imputations using linear mixed-effects model along with their packages are presented in Table S2.

4 | Multiple Imputation Diagnostics in Pharmacometrics

4.1 | Convergence

As briefly discussed before, monitoring the convergence of parameter estimates is crucial in evaluating the adequacy of the number of iterations in the MICE algorithm [15]. This is done by plotting one or more statistics of interest of M parallel

imputations against iteration number t via means of a line plot [19]. The statistics that are commonly plotted are the mean and standard deviation of the imputed data [19]. The imputation process is considered to have converged if the M lines seamlessly blend, displaying no discernible patterns [15]. In addition, the variance between lines should be no larger than the variance within each line [15]. This diagnostic plot can be generated in R using the `plot.mids()` function from the *mice* package [19].

4.2 | Plausibility of the Generated Imputations

Examining the concordance between observed and imputed data is a valuable method for evaluating the plausibility of imputations [47]. The rationale behind this approach is that high-quality imputations should be plausible [47]. Essentially, imputed values could have been the real ones if they had been observed [15]. Although there can be systemic differences between the two distributions under MAR scenarios, substantial deviations between imputed and observed data should raise concerns, as this suggests potential issues in the imputation process [47].

The *mice* package offers various graphical tools to assess the alignment of observed and imputed data, including `bwplot()`, `stripplot()`, `densityplot()`, and `xyplot()` [15]. The `stripplot()` function presents individual points for numerical variables per imputation, offering valuable insights into distributions, particularly in datasets with a limited number of data points [15]. For larger datasets, `bwplot()` is more suitable as this generates side-by-side box-and-whisker plots for observed and synthetic data [15]. Additionally, `densityplot()` displays kernel density estimates for both imputed and observed data [15].

4.3 | Pharmacometrics Model Evaluations

Performing MI adds complexity to model evaluations in pharmacometrics because it requires evaluating not just one model, but M multiplies imputed models. Standard tools such as goodness-of-fit plots, bootstrapping, and visual predictive checks remain applicable, but there are no established guidelines on how to apply them in the MI context. We propose evaluating the model with these tools using 10%–20% of the multiply imputed data sets selected randomly. Additionally, these tools can be applied to an averaged dataset derived from the M imputed datasets to assess the overall goodness-of-fit.

5 | Case Study: Warfarin Population Pharmacokinetics

We illustrated the superiority of MI compared to SI in estimating covariate effects in pharmacometrics when there is missing covariate data in a popPK case study of warfarin in healthy volunteers.

5.1 | Data Sources

We used a publicly available clinical dataset describing the pharmacokinetics of warfarin following a single-dose oral administration

in a total of 32 healthy volunteers [48]. The dataset contained 251 warfarin concentration measurements in total, with a median of six measurements per subject. Every subject was given a single dose of warfarin at $t=0$, and measurements were scheduled at 0.25, 0.5, 1.0, 2.0, 4.0, 6.0, 12.0, 24.0, 48.0, 72.0, 96.0, and 120.0 h post-dose. Body weight (BW) was fully observed for all subjects, with a median [interquartile range; IQR] of 71.7 [61.5–78.5] kg. Besides BW, sex and age were two complete subject characteristics included in the dataset. The dataset can be found in the Data S1.

5.2 | PopPK Model

A one-compartment popPK model parameterized with a first-order absorption rate constant (K_a), volume of distribution (V_d), and clearance (CL) was used to describe warfarin plasma concentrations. Random effects describe inter-individual variability (IIV) on CL and V_d . Residual variability was modeled using a combined additive and proportional error model. Baseline BW was a covariate on CL and V_d , modeled using a power function and centered around the median observed BW of 71.7 kg.

5.3 | Missing Data Generation

We used the `R produce_NA` function to generate five missingness scenarios, with 6.25%, 12.5%, 25%, 50%, and 75% BW data missing under a MAR mechanism conditioned on age and sex [49]. This corresponds to 2, 4, 8, 16, and 24 subjects with missing BW, respectively. Missing records are not subsets of one another. The high number of iterations is expected to reduce the impact of different occurrences of missingness across missingness scenarios, thereby guaranteeing comparability. We created 30 incomplete datasets for each missingness scenario (Figure 2).

5.4 | Data Imputation

For SI, we imputed missing BW in each incomplete dataset with the median observed BW from that dataset before fitting the popPK model (Figure 2).

For MI, our main imputation model was based on time, drug concentration, age, and sex. We also employed another imputation model as the secondary model where the predictors were the individual estimated CL and V_d , age, and sex, following a similar approach by Johansson et al. [1] The MI technique we used was PMM with the `R mice.impute.pmm` function [15]. We generated 20, 30, 45, 70, and 105 multiply imputed datasets to handle incomplete datasets with 6.25%, 12.5%, 25%, 50%, and 75% BW missing, respectively. This ensures that the number of converged models is at least equal to the percentage of data missing [19]. We then fitted a popPK model to each multiply imputed dataset, after which we pooled the parameter estimates of the converged models into one inference using Rubin's rule [19]. We did MI 30 times for each incomplete dataset (Figure 2) and then derived the median parameter estimate from these 30 imputed sets. All popPK analyses were performed in `nlmixr2` version 3.0.1, employing the first-order conditional estimation with the interaction estimation method [50]. A diagnostic plot showing the densities of the observed and multiply imputed BW

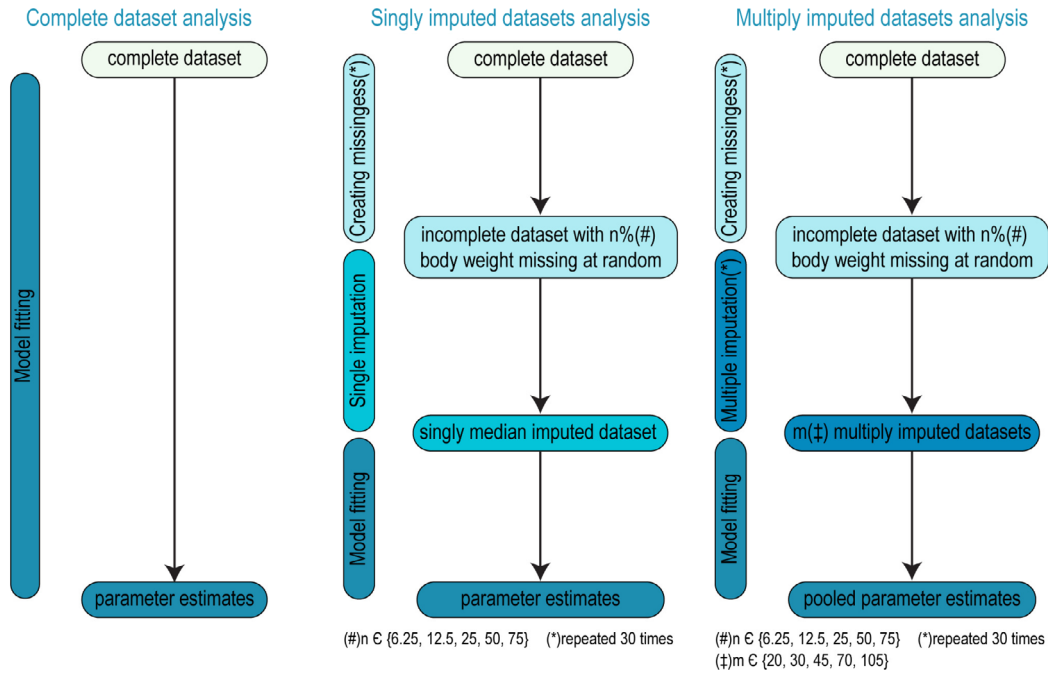


FIGURE 2 | The warfarin case study workflow.

in a scenario with 50% BW missing with our first MI approach is shown in Figure S1.

Example codes for SI and MI, along with package information, can be found in the Data S1.

5.5 | Performance Assessment

We assessed model stability for each imputation technique by calculating the convergence rate. We then compared the accuracy of covariate effect parameter point estimates from both SI and MI methods using the absolute relative bias of the estimate ($|rBias_estimate(\%)|$). Similarly, we calculated the absolute relative bias of the standard error ($|rBias_standard-error(\%)|$) to evaluate each technique's performance in accurately estimating uncertainty. The formulas for $|rBias_estimate(\%)|$ and $|rBias_standard-error(\%)|$ are as follows:

$$|rBias_estimate(\%)| = \left| \frac{CovariateEstimate_{SI/MI} - CovariateEstimate_{complete}}{CovariateEstimate_{complete}} \right| \times 100\% \quad (1)$$

$$|rBias_standard-error(\%)| = \left| \frac{StandardError_{SI/MI} - StandardError_{complete}}{StandardError_{complete}} \right| \times 100\% \quad (2)$$

where $CovariateEstimate/StandardError_{SI/MI}$ refers to the covariate effect estimate/standard error obtained from the incomplete dataset treated with SI or MI and $CovariateEstimate/StandardError_{complete}$ refers to the estimate/standard error from the complete dataset.

For each missingness scenario, we compared $|rBias_estimate(\%)|$ and $|rBias_standard-error(\%)|$ obtained from SI and MI using a paired one-sided Wilcoxon rank-sum test for two dependent, non-normally distributed samples [51]. Additionally, we compared the variances of the $|rBias_standard-error(\%)|$ obtained with the two methods using the Fligner-Killeen test [52]. A 5% significance level was chosen for these tests.

Similar comparisons for the structural PK parameters (i.e., K_a , V_d , and CL) were performed.

5.6 | Results

The convergence rates of both imputation techniques were 100% across the missingness scenarios.

We presented the $|rBias_estimate(\%)|$ and $|rBias_standard-error(\%)|$ for SI and MI regarding the covariate effects across all five missingness scenarios using the main imputation model, along with the corresponding p -values, in Figures 3 and 4, respectively. Similar results for the structural PK parameters are presented in Figures S2 and S3. The analysis results obtained from the secondary imputation model are shown in Figures S4–S7. Overall, the figures showed an increasing magnitude of $|rBias_estimate(\%)|$ and $|rBias_standard-error(\%)|$ for all parameters as BW missingness increases. The magnitude of $|rBias_estimate(\%)|$ for the structural PK parameters was minimal (15% at most for the 75% missingness case), with the smallest for K_a , and smaller than that of the covariate effect parameters (Figures 3, S2, S4 and S6). Similarly, $|rBias_standard-error(\%)|$ of the structural PK parameters was also smaller than that of the covariate effect parameters in general, with the figure for K_a being the smallest (Figures 4, S3, S5 and S7).

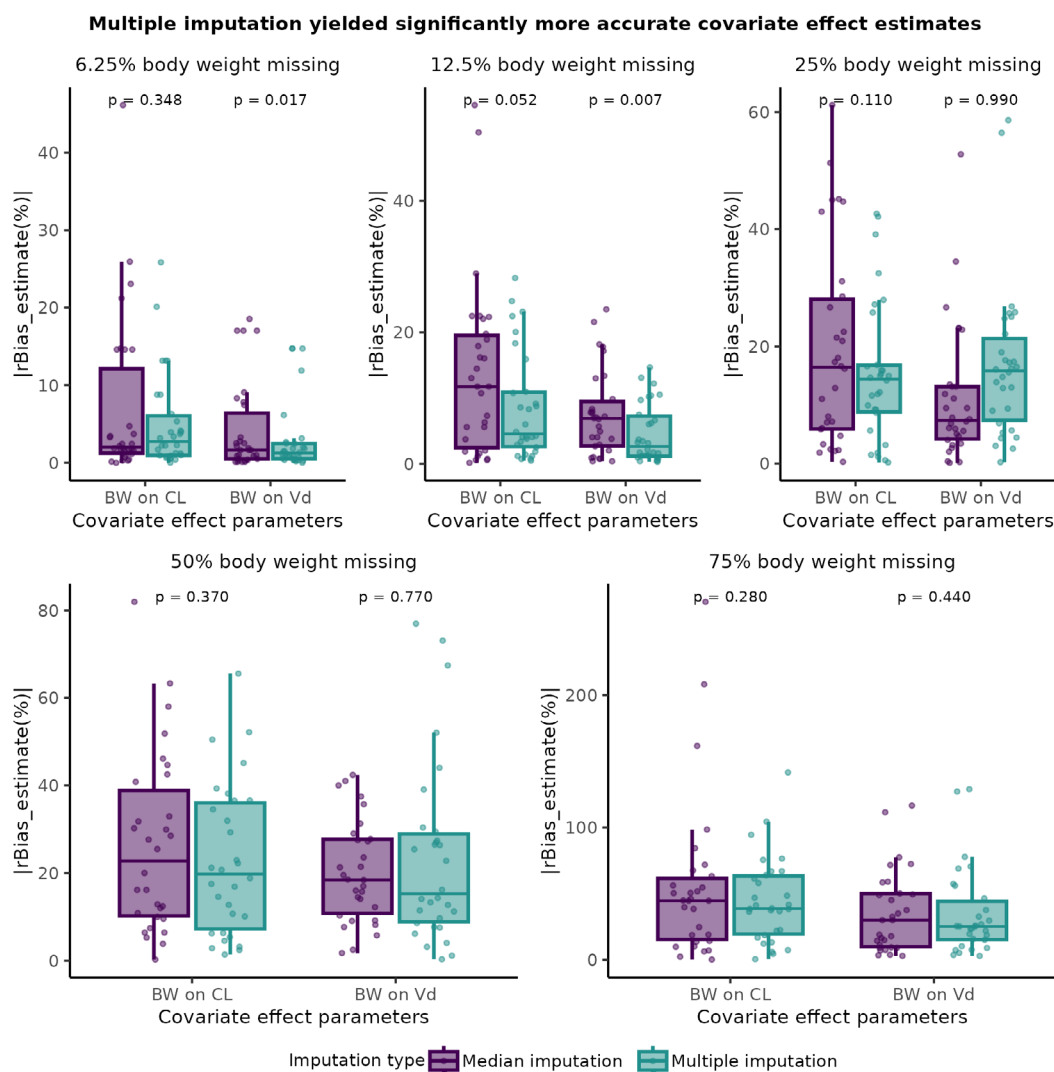


FIGURE 3 | The absolute relative bias of the covariate effect estimate (%) obtained from median imputation and multiple imputation using the main imputation model in five missingness scenarios. $|rBias_estimate(\%)|$: Absolute relative bias of the estimate (%).

In estimating the covariate effects, MI produced more accurate estimates, with significantly lower $|rBias_estimate(\%)|$ for the effect of BW on V_d in the 6.25% and 12.5% missingness scenarios. The respective p -values were 0.017 and 0.007 (Figure 3).

Similarly, MI estimated the uncertainty of covariate effect estimates significantly better than SI overall. Specifically, MI reflected the uncertainty estimate better for the effect of BW on V_d in the 6.25%, 25%, and 50% missingness scenarios and for the effect of BW on CL in the 6.25% missingness case. The corresponding p -values were 0.024, 0.048, 0.007, and 0.040 (Figure 4). In addition, the Fligner-Killeen test showed that the $|rBias_standard-error(\%)|$ obtained with MI had significantly lower variance than those obtained with SI in two cases: the 25% missingness case for the effect of BW on V_d and the 50% missingness case for the effect of BW on CL. The respective p -values were 0.0029 and 0.030.

The superior performance of MI in accurately estimating the point estimate was again demonstrated for the structural PK parameters, with significant superiority in almost all missingness scenarios (Figure S2). Additionally, MI's benefit was shown to be more evident in reflecting the uncertainty estimate, as

significant superiority was observed in more missingness scenarios (Figure S3).

The superiority of MI over SI in accurately estimating the point estimate for all five parameters was still evident when using the secondary imputation model (Figures S4 and S6).

However, MI's performance in accurately reflecting the parameter uncertainty was significantly worse in the 50% and 75% missingness scenarios (Figures S5 and S7).

6 | Discussion

Missing covariate data occurs frequently in medical research, including pharmacometrics. Appropriately handling covariate missingness is crucial and is emphasized by both the FDA and EMA [4, 53, 54]. The ICH E9 guidance on statistical principles for clinical trials encourages the use of imputation techniques to minimize the missing data problem, thereby mitigating the need to use different analysis sets [4]. However, this guidance recommends the LOCF technique, which is highly prone to

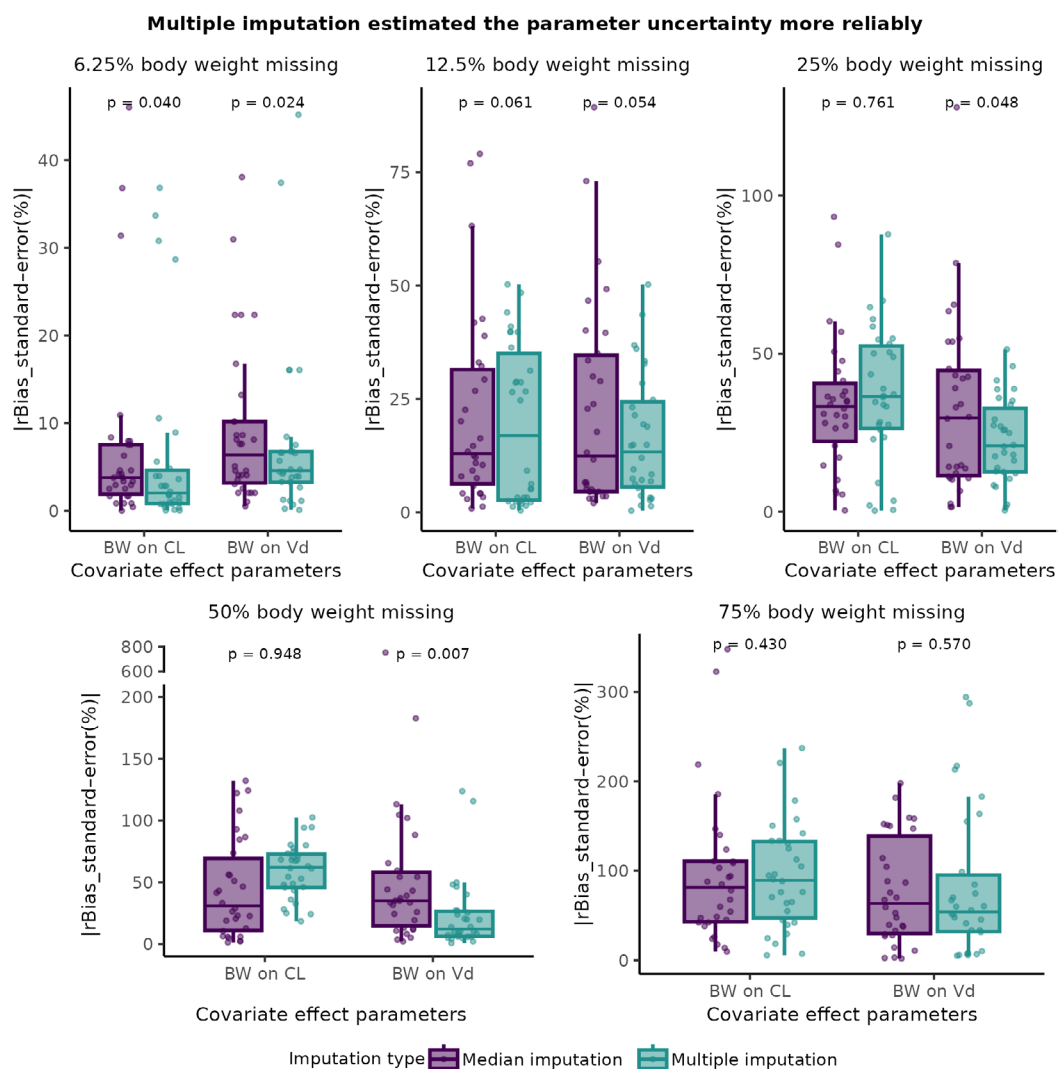


FIGURE 4 | The absolute relative bias of the covariate effect standard error (%) obtained from median imputation and multiple imputation using the main imputation model in five missingness scenarios. $|rBias_standard-error(\%)|$: Absolute relative bias of the standard error (%).

bias. In pharmacometrics, handling covariate missingness has also not yet been optimal, as pharmacometricians commonly employ SI, a technique that replaces the missing covariate with a single value and treats it as if it were true. In contrast, MI replaces the missing values multiple times, better reflecting the uncertainty about the missing data, making it a more reliable approach. However, the application of MI remains limited, although its usage has been growing recently [1, 2, 55–59]. We have therefore provided a brief introduction to MI and an MI workflow in pharmacometrics modeling, encouraging a broader adoption of MI in the field. We illustrated the application of MI and its superiority over SI through the analysis of warfarin pharmacokinetics data in healthy volunteers [10].

Our study is not the first to present an MI workflow in pharmacometrics, but it is the most comprehensive. An MI workflow was initially proposed by Wu and Wu in 2001 and later expanded by Johansson in 2013, who implemented it as part of the *mimp* functionality in PsN [1, 55]. However, these papers did not address important topics, such as the optimal number of imputations or ensuring the normality assumption of parameters before applying Rubin's rule.

The *mimp* MI workflow differs from the MICE workflow presented in our paper in several ways. First, *mimp* uses individual parameter estimates as predictors for partially missing covariates, rather than directly using the outcome variable as we did [1]. This method, while different, similarly captures the information contained in the outcome variable. However, *mimp* cannot handle multivariable missing covariates—an issue that often arises in practice. In *mimp*, each missing covariate is imputed individually, whereas the MICE algorithm in the R *mice* package imputes multivariable missing data simultaneously [1].

Moreover, the *mimp* workflow only uses fully observed covariates to predict partially missing ones, meaning that covariates with partial missingness cannot serve as predictors [1]. As a result, when multiple covariates have missing values, *mimp* cannot fully leverage all available data, potentially biasing covariate estimates. In contrast, *mice* utilize all available information, whether fully or partially observed, reducing the likelihood of bias.

Additionally, implementing *mimp* is more complex, as it requires separate model files for each step of the workflow [60]. With *mice*, the entire workflow can be executed in a single

function. On the other hand, *mimp* allows pharmacometricians to perform MI using NONMEM, which may be more convenient if NONMEM is already used for model building. With *mice*, pharmacometricians must use R for MI and thus might benefit from using an R-based estimation tool like *nlmixr2* for the full pharmacometrics workflow. Alternatively, tools like *pharmr* allow estimating model parameters with NONMEM from within R, facilitating the implementation of *mice* in NONMEM [61].

We used the FCS algorithm for doing MI, which has the main advantage of not having to pre-specify a sequence in which covariates are imputed, thus ruling out that there might be any impact of the chosen sequence [8]. To what extent the results of alternative approaches are impacted by the choice of sequence of imputation would require an extensive, dedicated simulation study, which is outside of the scope of our work.

Our brief introduction to MI is not exhaustive. Readers interested in delving deeper into the topic can refer to several excellent texts [8, 15, 35]. We focused specifically on MI in studies with missing baseline covariates, where no hierarchical structure is present. For a more detailed discussion on methods for applying MI to hierarchical or repeated measurement data, readers can refer to other works [45, 62–64]. Similarly, we emphasized parametric methods (e.g., logistic regression or linear regression models) for building imputation models, though machine-learning methods for MI have also been explored [65]. Additionally, we focused on MI under the assumption that the data are MAR. The MI methods described can be modified for cases where data are believed to be MNAR. In such cases, sensitivity analyses are necessary, with three broad approaches available: selection modeling, latent variable modeling, and pattern mixture modeling [35]. Carpenter et al. provide a summary of various methods for addressing MNAR data [35].

We have focused on the MICE algorithm to perform MI for multivariate missing covariates, but this is not the only approach. Another common approach is “joint modeling”, where variables in the imputation model are typically assumed to follow a multivariate normal distribution [66]. After estimating the parameters of this distribution, missing values can be imputed by randomly drawing from it. In theory, this method assumes all variables are continuous, making it less suitable for categorical covariates (e.g., biological sex). Irby et al. recently demonstrated the superiority of the joint modeling approach over complete case analysis in identifying continuous covariate effects in pharmacometrics [57]. However, some studies have shown that multivariate normal joint modeling performs comparably to MICE, even when categorical variables are included [66, 67]. Conversely, some research indicates that joint modeling with noncontinuous variables may not perform as well [8]. The MICE algorithm remains an attractive option due to its flexibility and ability to handle different types of variables.

In our case study, the magnitudes of $|\text{rBias_estimate}(\%)|$ and $|\text{rBias_standard-error}(\%)|$ increased with higher BW missingness. This is as expected since the uncertainty around the missing values grows with increasing missingness. Additionally, structural PK parameters had smaller $|\text{rBias_estimate}(\%)|$

and $|\text{rBias_standard-error}(\%)|$ compared to covariate effect parameters. This is also expected, as covariate missingness influences the covariate effect estimates more than those of the structural PK parameters. Among the structural PK parameters, K_a had the smallest $|\text{rBias_estimate}(\%)|$ and $|\text{rBias_standard-error}(\%)|$. This is because BW was not a covariate on K_a ; hence, BW missingness impacts K_a minimally.

Multiple imputation produced more accurate covariate parameter estimates than SI overall. This underscores the advantages of MI and the harmful bias introduced by SI, which aligns with a previous study in pharmacometrics [2]. Although SI is easier to implement, its use can lead to detrimental consequences, and convenience should not justify neglecting MI. The improvements in absolute bias with MI in our study were not always statistically significant due to the small sample size (32), the issue that was highlighted in earlier research [68]. However, our case study demonstrates that MI can still offer benefits regarding bias even in small sample datasets commonly encountered in pharmacometrics. In addition, our case study is the first in pharmacometrics to demonstrate the superiority of MI over SI in reducing absolute bias across various levels of missingness, highlighting MI's great potential for handling missing covariates.

The superiority of MI over SI in our case study was demonstrated to be more profound in correctly reflecting the uncertainty of all parameter estimates, including covariate effect estimates, across all missingness scenarios. This highlights the major benefit of MI over SI, as the former acknowledges the uncertainty of the missing values and hence reflects the parameter uncertainty better than the latter. Our finding aligns with a previous study of 215 neonates, where 20% of the covariate was MAR. They confirmed that MI yielded a standard error estimate of the covariate effect significantly closer to the truth [3]. Our case study is the first in pharmacometrics to compare the absolute relative bias in estimating the standard errors between MI and SI under various degrees of covariate missingness.

In our case study, the MI performance when using the main imputation model was better than when using the secondary imputation model, especially in reflecting the uncertainty estimate. This may be because the individual parameter estimate lacks information due to eta-shrinkage.

With our case study, we have only investigated the scenario where there is only one covariate missing and the missing covariate is at baseline. Future research should explore MI's performance when two or more covariates are missing and also where at least one covariate is time-varying (e.g., SCr). In addition to MI, maximum likelihood (ML) is another valid approach under MAR if one treats the covariates as outcomes and models them jointly with the outcome. Such an approach has been implemented in the full random effects model (FREM) to handle MAR covariates [69]. However, this FREM technique is a multivariate longitudinal mixed model, which encounters many challenges in model fitting [70]. Nevertheless, a comparison between FREM and MI would provide a more comprehensive understanding of their applications in pharmacometrics. This is beyond the scope of the current work but remains an important direction for future research.

Computer runtime may be an issue that hinders the adoption of MI in practice. The runtime is largely determined by the extent of missing values, with more imputations needed if the missingness increases, and hence the runtime. In our case study, the runtime for one MI procedure ranged from 4 to 21 min, corresponding to the BW missingness of 6.25% and 75%. Each task was run on one core in one big memory node. Each node contains 2 Intel Xeon Platinum 8360Y CPUs@2.4GHz (Ice lake), with 36 cores each (2 NUMA domains and 1 L3 cache per CPU), and the default memory per core is 28,000 MB.

In summary, MI is a preferable approach in handling covariate missingness under MCAR and MAR. We demonstrated that MI generally leads to more accurate covariate effect point estimates and reflects the uncertainty of covariate effect estimates better than SI, regardless of the degree of missingness. We encourage pharmacometricians to adopt MI due to its clear advantages over the historic SI method.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.